

Intro to HPC

- COL774

Setting up the Account

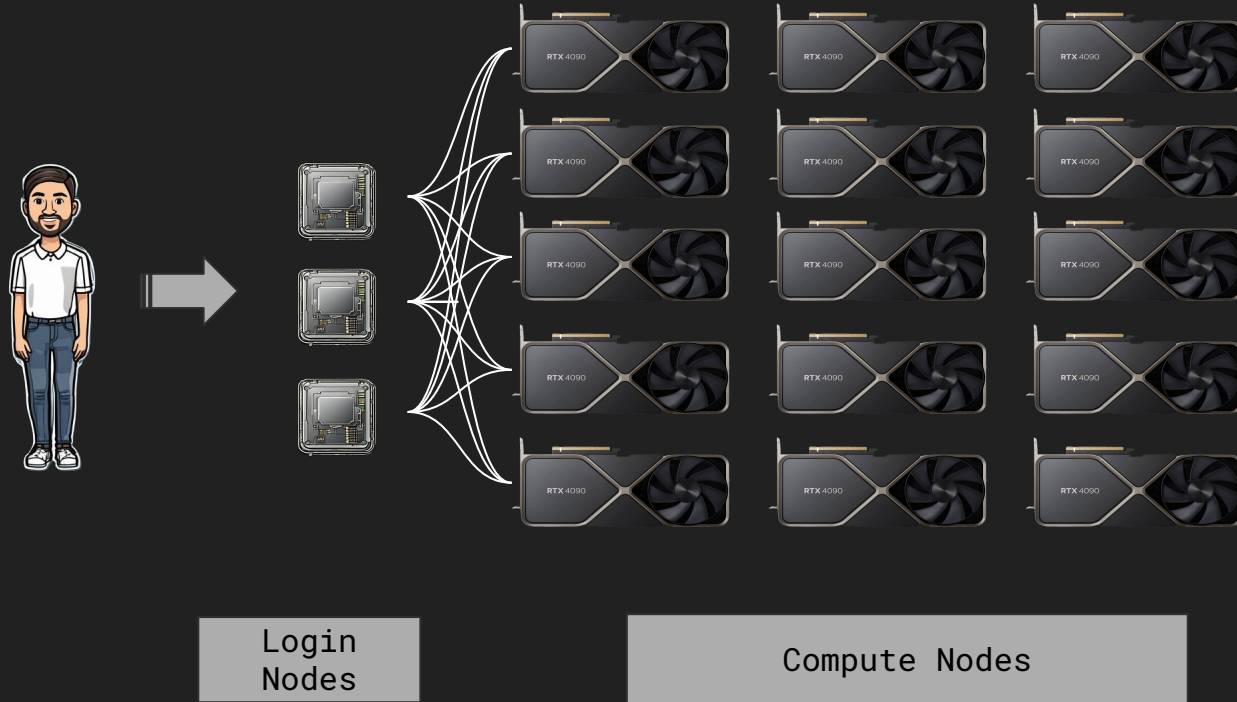
- Logging in:

```
ssh <kerberos_id>@hpc.iitd.ac.in  
Eg. ssh aiz228170@hpc.iitd.ac.in
```

- To have a hassle free journey, users are expected to run a one-time script that sets up proper permissions for the HPC account.

```
source /home/apps/skeleton/oneTimeHPCAccountEnvSetup.sh
```

HPC Structure



Internet Access : (Check Usage at track.iitd.ac.in)

a. On your laptop: (One time)

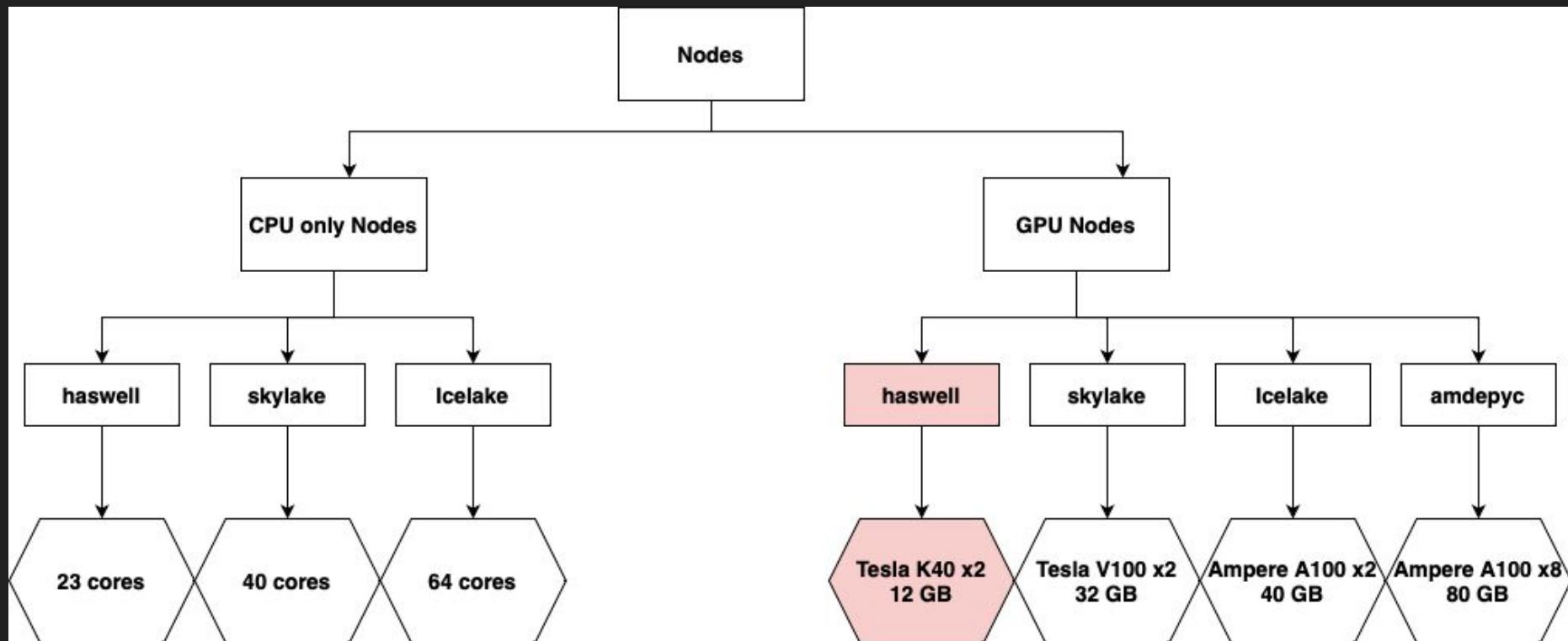
- `git clone https://github.com/kanha95/HPC-IIT-Delhi.git`
- `edit proxyAuth.txt`
- download certificate from <http://www.cc.iitd.ernet.in/certs/CCIITD-CA.crt> and place in the directory.
- `scp -r ./HPC-IIT-DELHI <kerberos_id>@hpc.iitd.ac.in:~/`

b. On a new terminal:(Repetitive)

- `ssh <kerberos_id>@hpc.iitd.ac.in`
- `cd HPC-IIT-DELHI`
- `export SSL_CERT_FILE=$PWD/CCIITD-CA.crt`
- `python2 iitdproxy.py proxyAuth.txt`

Installing Anaconda

- On a new terminal again
 - `export http_proxy="http://10.10.78.XX:3128"`
 - `export https_proxy="http://10.10.78.XX:3128"`
 - Note XX: is the degree id (replace xx with your degree)
 - `{'btech':22,'dual':62,'diit':21,'faculty':82,'integrated':21,'mtech':62,'phd':61,'retfaculty':82,'staff':21,'irdstaff':21,'mba':21,'mdes':21,'msc':21,'msr':21,'pgdip':21}`
- `wget "https://repo.anaconda.com/archive/Anaconda3-2023.09-0-Linux-x86_64.sh"`
- `bash ./Anaconda3-2020.05-Linux-x86_64.sh`
- `conda init`
- `conda config --set auto_activate_base True`



Job Submission

```
qsub -P <project_id> -q <standard/low/high> -lselect=1:ncpus=x:ngpus=y -lwalltime=10:00:00 -I
```

- Haswell / Skylake nodes:
 - `qsub -P col774 -q standard -lselect=1:ncpus=4:ngpus=1:centos=haswell/skylake -lwalltime=168:00:00 -I`
- Icelake nodes:
 - `qsub -P col774 -q test -lselect=1:ncpus=4:ngpus=1 -lwalltime=168:00:00 -I`
- Amdepyc nodes:
 - `qsub -P <department project> -q scai_lowq -lselect=1:ncpus=4:ngpus=1 -lwalltime=24:00:00 -I`

Utility Commands

- `qdel <job_id>` : Delete a job from queue.
- `qstat -u $USER` : List all the jobs under user.
- `qstat -f <job_id>` : Shows status of an active job.

Storages - (`lfs quota -hu $USER /home`)

- `$HOME`: 100 GB → Permanent Storage
- `$SCRATCH`: 25TB → Temporary : If storage becomes low data may be deleted without prior notice.
- Use `$HOME` for storing codes etc.
- Use `$SCRATCH` for model checkpoints, datasets etc.

Working on Vscode + Jupyter-notebooks

Demo!!!

<https://supercomputing.iitd.ac.in/?home>

https://pytorch.org/tutorials/beginner/basics/quickstart_tutorial.html#loading-models